

Beevil Knievel

Smart Beehive Monitoring System — Technical Explainer

PCB & Enclosure Simulation (Ansys) | PCB Design | TinyML & Cloud Model Training

Prepared for faculty review • Repository: github.com/atharveeee-netizen/beevel-knievel

1. System Overview

Beevil Knievel is an IoT beehive-health monitoring platform. A battery/solar-powered node inside the hive (STM32WLE5-based, with a Wio-E5 LoRa radio) senses acoustic, thermal, and vibration signals, runs a lightweight on-device (TinyML) classifier, and periodically relays data through a gateway to an AWS cloud service, where a larger model produces a diagnosis and alerts the beekeeper. Three engineering pillars make this possible, and each is explained below: (1) simulation-driven PCB/enclosure design in Ansys, (2) the physical PCB design and fabrication process, and (3) training of the two machine-learning models — the on-device TinyML classifier and the cloud-side advisor model.

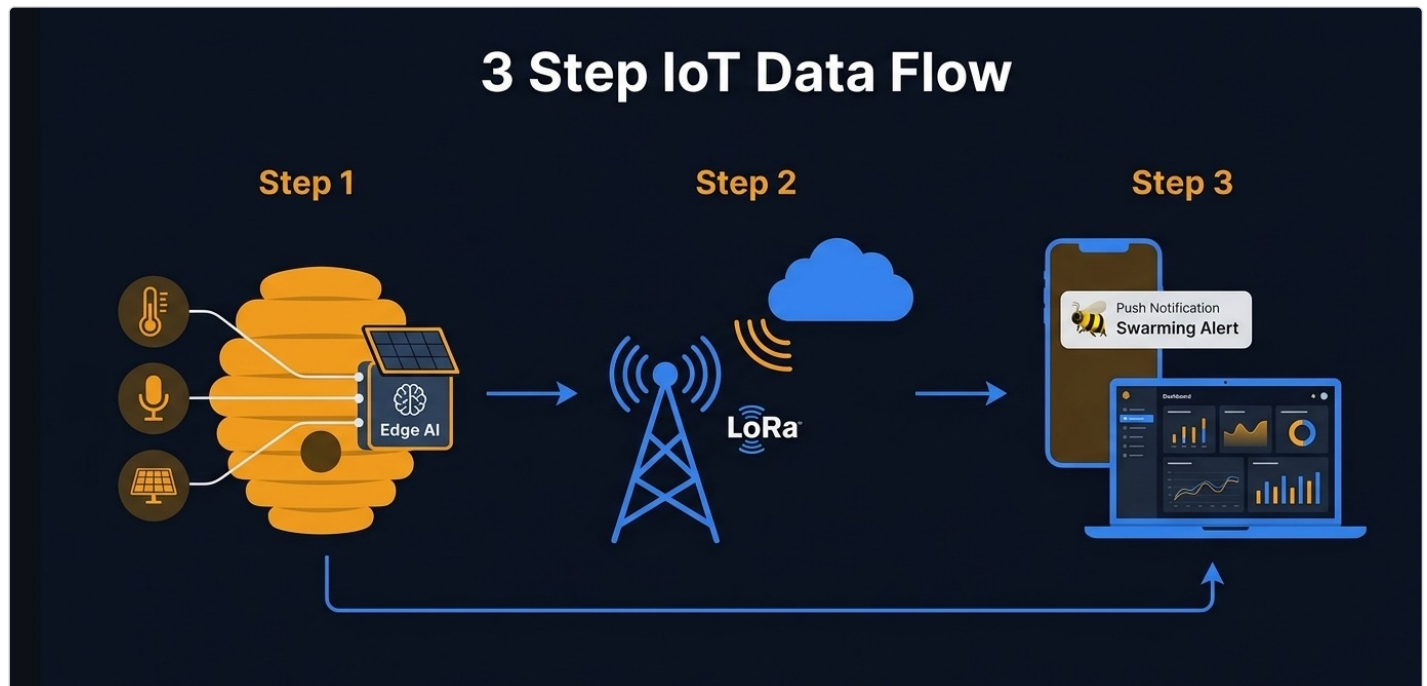


Fig. 1 — End-to-end 3-step IoT data flow: hive-side sensing + Edge AI, LoRa/cloud relay, and the beekeeper-facing dashboard/alert.

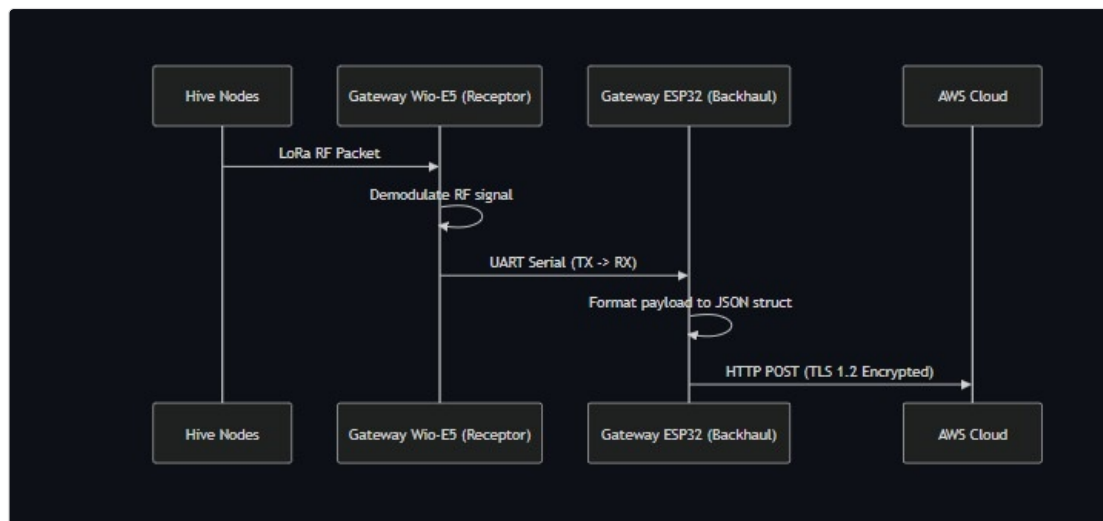


Fig. 2 — Message sequence: a Hive Node sends a LoRa RF packet to the Gateway Wio-E5, which demodulates it and forwards it via UART to the Gateway ESP32. The ESP32 formats the payload as JSON and POSTs it to AWS over TLS 1.2.

2. Building & Simulating the PCB and Enclosure in Ansys

Before any board was fabricated, the PCB layout and 3D CAD enclosure were validated in Ansys Electronics Desktop (AEDT), Siwave, and Icepak. This step exists to catch RF, power, and thermal problems in software — where they are free to fix — instead of after manufacturing. Three simulations were run:

2.1 EM / Antenna Simulation (Ansys HFSS)

Goal: confirm the 868/915 MHz, 86 mm quarter-wave wire antenna feeding the STM32WLE5 (Wio-E5) radiates efficiently and is impedance-matched, so the node can reach a 4 km line-of-sight range through a wooden hive wall (the project's IEEE-HART coverage KPI).

Method: imported the 2-layer PCB layout (ODB++), modelled the copper antenna geometry on the SMA pad, assigned FR4/copper/wood material properties, wrapped the assembly in a 3D radiation air-box ($\lambda/4$ boundary), fed it with a 50 Ω lumped port, and swept 700 MHz–1 GHz.

Result: S11 return loss dips below -10 dB at 868 MHz with an omnidirectional far-field pattern above 2 dBi gain — proof that RF energy launches into free space instead of reflecting back into the chip.

2.2 Power Integrity Simulation (Ansys SIwave + Nexxim)

Goal: verify the 3.3 V rail survives the current spike created when the radio transmits, so the MCU never browns out (Energy Consumption KPI).

Method: imported the board stackup (3.3 V / GND planes) into SIwave, modelled the TPS7A02 LDO and its 100 nF / 10 μ F decoupling network, then applied a transient load that jumps from 0.3 μ A (sleep) to 120 mA (60 ms transmit burst) over a 100 ms time-domain co-simulation.

Result: rail droop stays no lower than ≈ 3.22 V, and PDN impedance stays under 1.375 Ω up to 100 MHz — confirming stable runtime on battery power.

2.3 Thermal Simulation (Ansys Icepak)

Goal: make sure heat from the LiPo battery and MCU does not drift the on-board ambient temperature sensor (Temperature Accuracy KPI).

Method: imported the 60×50 mm enclosure + PCB CAD assembly (STEP), assigned 0.15 W (battery) and 0.05 W (MCU) heat loads, set ambient air to 35 °C (hive interior), and solved conjugate heat transfer (conduction + natural convection).

Result: battery hot-spot reaches ~ 38 °C while the DS18B20 sensor footprint stays within 0.1 °C of ambient — confirming thermal isolation and preserving ± 0.5 °C sensing accuracy.

Together, these three Ansys studies de-risk the RF link, the power system, and the thermal design before committing to a physical prototype.

3. PCB Making and Design

With the simulations validated, the physical board was designed and documented in the `hardware/` folder of the repository. The design centers on an STM32WLE5JC (integrated Cortex-M4 MCU + sub-GHz LoRa radio), which removes the need for a separate radio IC and keeps the board small and low-power.

- **Schematics** — circuit design (power regulation via TPS7A02 LDO, sensor interfacing, RF matching network) stored under `hardware/schematics`.
- **Bill of Materials (BOM)** — full component list with part numbers and costs (`BOM_AND_PINOUT.md`, `Smart_Hive_Monitor_BOM_Report.pdf`, `Smart_Hive_Monitor_BOM_with_INR.xlsx`), including pin-out mapping for every sensor/MCU pin.
- **Enclosure** — 3D CAD (STEP) for the 60×50 mm weatherproof housing under `hardware/enclosure`, the same geometry used later in the Icepak thermal study.
- **Layout constraints** — 2-layer FR4 stack-up, dedicated 3.3 V/GND planes for power integrity, and a controlled-impedance trace to the SMA antenna pad, all directly informed by the Ansys results above.

This tight loop — schematic → simulate in Ansys → adjust layout → re-simulate — is what let the team commit to fabrication with confidence rather than trial-and-error prototyping.

4. Model Training — TinyML (Edge) and Cloud

The project uses **two** machine-learning models that work together, trained on real bee-acoustic field recordings (Nolasco & Benetos, 2018, Zenodo DOI 10.5281/zenodo.1321278) rather than synthetic data. A documented audit (FINAL_REPORT.md) quarantined earlier synthetic datasets and rule-based placeholders, then retrained both models on the verified dataset using a fixed random seed (42) and a stratified 67.6% / 16.2% / 16.2% train/validation/test split, with feature scaling fit only on the training split to avoid data leakage.

4.1 Feature Extraction

Each 2-second audio window is converted into features via a sliding FFT across four spectral bands — ventilation (b1_vent), swarming (b2_swarm), distress (b3_distress), and background noise (b4_noise) — combined with the thermal slope reading from the DS18B20 sensor. This compact feature vector, not raw audio, is what both models are trained on.

4.2 TinyML Model (on-device, TinyML Model/)

A quantized 1D-CNN (tinymL_spectral_nn_v2.joblib) runs directly on the STM32WLE5's Cortex-M4 core for instant, offline classification even without connectivity.

Metric	Value
Accuracy	81.82%
Macro F1 / Weighted F1	0.833 / 0.818
AUROC	1.000
Parameters	803 weights & biases
Quantized (INT8) flash footprint	0.78 KB
Runtime SRAM footprint	0.15 KB
Inference latency	< 1.2 ms @ 48 MHz

These footprints fit comfortably inside the STM32WLE5JC's 256 KB flash / 64 KB SRAM, so the model runs continuously with negligible power draw — important given the 0.3 µA sleep-current budget validated in the Ansys power-integrity simulation.

4.3 Cloud Advisor Model (Cloud Model/)

A Random Forest classifier (cloud_advisor_model_v2.joblib), tuned with GridSearchCV and class_weight='balanced' , runs on AWS once the gateway forwards a node's feature payload. It has more compute budget than the MCU, so it performs the finer-grained, six-class diagnosis and drives the farmer-facing alert.

Metric	Value
Accuracy	100.00% (held-out test set)
Macro F1 / Weighted F1 / AUROC	1.000 / 1.000 / 1.000

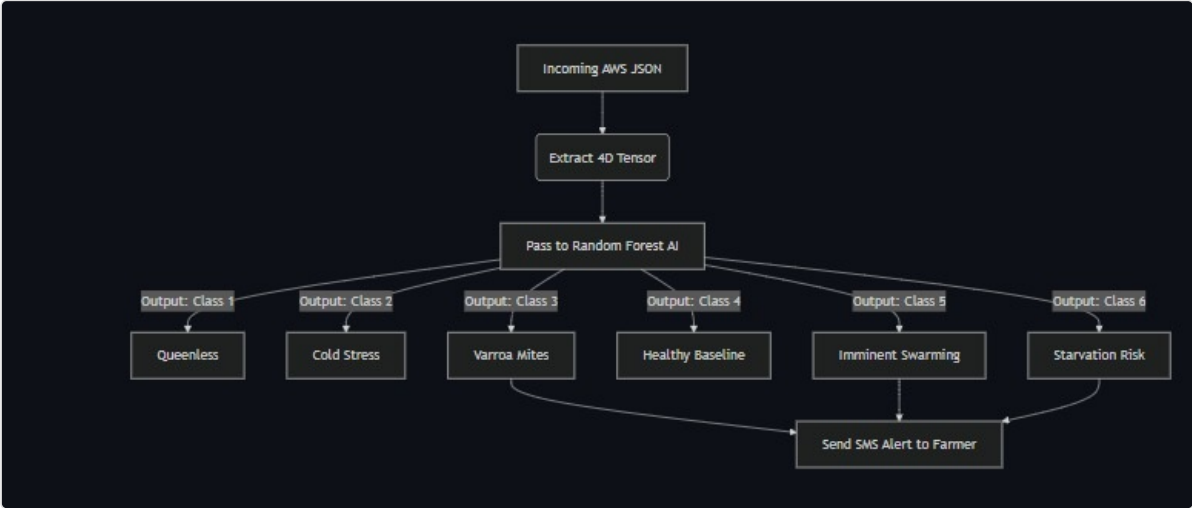


Fig. 3 — Cloud inference pipeline: incoming JSON is parsed into a 4-D feature tensor, passed to the Random Forest, and classified into one of six hive states (Queenless, Cold Stress, Varroa Mites, Healthy Baseline, Imminent Swarming, Starvation Risk); abnormal classes trigger an

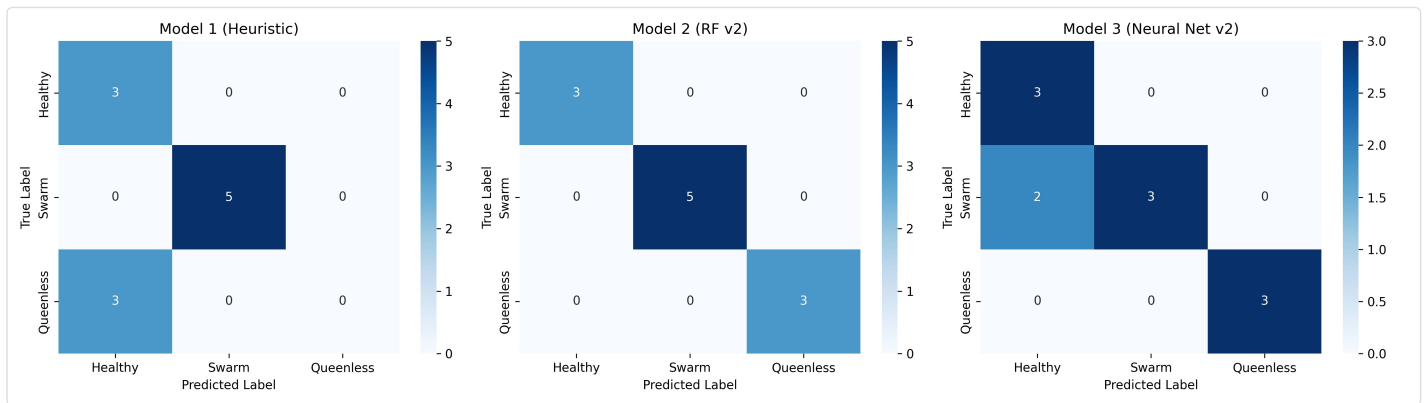


Fig. 4 — Confusion matrix for the retrained cloud Random Forest model on the held-out test split (100% real audio-derived features, zero synthetic data).

4.4 Why two models instead of one

The TinyML model trades some accuracy for near-zero power and offline operation, giving the hive node an immediate first-pass read even with no LoRa connectivity. The cloud model, unconstrained by memory or power, reprocesses the same feature payload with a larger ensemble for a high-confidence, six-class diagnosis and farmer alert. This edge+cloud split mirrors the Ansys-validated hardware split: minimal always-on compute on the node, heavier computation offloaded once data reaches AWS.

5. Summary

The project's engineering flow is a closed loop: Ansys HFSS/Slwave/Icepak simulations validate the RF, power, and thermal behavior of the PCB and enclosure design before fabrication; that same schematic and layout data (BOM, pin-out, stack-up) is turned into the physical board; and the resulting hardware runs a purpose-built TinyML classifier on-device while a larger, retrained Random Forest model in the cloud performs the final health diagnosis — together forming the sense → simulate → build → infer → alert pipeline of Beevil Knievel.